

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Potometer	1			1		1
		(ii)		(Leafy shoot must be) cut underwater. (1) {Apparatus / description of} must be assembled underwater / potometer flooded with water. (1)	2			2		2
		(iii)		Any two for one mark Temperature / light <u>intensity</u> / humidity		1		1		1
	(b)	(i)	I	157 / 157.08 / 157.1 (2 marks) Award 1 mark for (3.14 or π) $\times 0.5^2 \times 200$ 628 / 628.3 (use of diameter instead of radius)		2		2	2	2
			II	49.1 (3 marks) If incorrect award 2 marks for 49 / 49.06 / 49.09 (correct answer but not to 1 dp) If incorrect award 1 mark for 157 (answer from (b)(i) / 192×60 Allow ecf from (b)(i)		3		3	3	3
		(ii)		Any three (x1) from A. As fan distance increases the <u>rate</u> of water uptake decreases (1) B. As distance increases transpiration (rate) decreases (1) C. Water vapour accumulates / humidity increases / diffusion shells remain outside of the leaf (1) OWTTE D. Decreases the steepness of the {diffusion / water potential} gradient (1) ignore concentration Accept reverse argument for each marking point		3		3		3